

MAKE ROOM FOR THE CADDY

Since homes continue to be built with attached garages—with no change in sight—builders and contractors need to understand how to minimize the risk of CO poisoning.

BY PAUL H. RAYMER

Most homeowners would object to dedicating a room in their house to a cow. Yet the majority of houses in this country are constructed with a room dedicated to shelter a far more smelly and deadly beast: the family car. Lawsuits related to attached garages abound. The separation between the place where the people live and the place where the car is stored must be much more clearly defined.

In colonial times, houses were often built over a shelter for animals—the first tuck-under. The animals added warmth to the home. Having to put up with the animal odor was a small price to pay. With the advent of the automobile in the 1890s, and of the attached garage in the 1930s, convenience far outweighed the problem of gases and odors from the car. There was enough natural ventilation in homes to remove the harmful gases.

Even as houses became tighter, there was an intuitive sense among people that the garage should be considered and treated as exterior space. This made it an acceptable place to store the car, the lawn mower, extra paint, gasoline, and other toxic substances. It was a place that could be used for grilling steaks on rainy days, a place for working on projects, spray painting the lawn furniture, and venting the dryer. The separation between the house and the garage didn't need to be treated with quite as much care as other parts of the building envelope because the garage wasn't really outside and wasn't really inside either.



Sealing the house/garage (HG) interface with a tuck-under garage, pictured above, can be a particularly difficult task due to the indirect connections through the ceiling and the wall structures above.

But with the growing concern about indoor air quality (IAQ) and green building, more research has been done on the transfer of pollutants from the garage to the house. For me, this research clearly demonstrates that more attention needs to be paid to reducing or eliminating this flow of poisonous gases.

How Serious Is the Problem?

Of the more than one million houses built in the United States in 2003, 93% included a one-, two-, or three-car garage (U.S. Bureau of the Census). That percentage has remained fairly stable for the past five years.

In 2002, the Alaska Building Science Network monitored 65 homes for car-

bon monoxide in 4 Alaskan cities. For most of these homes, CO readings mimicked the pattern of the garage CO readings. In 1996, Professor Tom Greiner of the University of Iowa carefully analyzed an Iowa home where one of the occupants was suffering from regular headaches. Using pressure measurements, smoke tests, and tracer gas, Greiner found that the startup of the owner's car was the only source of CO.

A 2001 study of benzene from automobiles in attached garages in the United Kingdom showed that, although benzene levels were much lower in the houses than in the garages, the average level was still at least several times higher than outdoor concentrations. Minnegasco, a natural gas utility in Minnesota, responded to more than 16,000

CO calls in 1996. They commissioned a study to try to determine the source of the problem in houses where two or more complaints had been investigated and where the technicians were unable to identify the source of the CO. In 74% of the houses studied, CO was carried into the house with infiltration from attached and tuck-under garages.

In all of these studies, the problem was essentially one of air movement. As long as air can move easily through the house/garage (HG) interface, the pollutants will be carried along. If the pressure in the house is lower than the pressure in the garage, the flow of air will be from the garage to the house. Any negative pressure difference in the garage relative to the house will move the air through small holes in the HG interface. With a small pressure differential, the leakage from the garage to the house is slow; on average, peak pollutant levels do not appear in the house until an hour after the car has left the garage. For this reason it has been difficult to see the one-to-one connection between starting the car in the garage and the CO alarms and occupants' health symptoms in the house. The elevated level of CO in the garage can continue for four to six hours after a car leaves the garage, and that air infiltrates the house the entire time.

Mild CO poisoning feels like the flu. More serious CO poisoning can lead to breathing difficulties, lack of reasoning, and death. Like most other IAQ issues, different people are affected differently by CO. It is most likely to have the most serious effects on the very young, the sick, and the elderly. Even levels of CO previously thought to be extremely low have been shown to lower the birth weight of children whose mothers were exposed to CO during the last trimester of their pregnancies.

The problem may be more serious in garages in colder climates, because the car may be allowed to run longer in the garage to warm the engine before venturing out, and the house may be more carefully sealed with less naturally driven ventilation. Also, cold engine exhaust gases are dirtier, and the engine runs longer in this high-emission mode in cold weather. But with all the reports of lawsuits coming from warm climates, it's

clear that there is no way to guarantee the safety of occupants in a house with an integral garage where a car is allowed to run for an extended period of time—in any part of the country.

Not Just CO

When a car starts, it can send levels of CO soaring to 80,000 ppm; however, CO is not the only pollutant transferred from an attached garage. As the car cools, it expels an array of other fumes. An EPA test to measure the fuel evaporation from a vehicle in a chamber (called the hot soak test) requires the vehicle to be run for 41 minutes, after which the evaporation emissions are measured. This provides a measurement of the total hydrocarbons or THC. Not surprisingly, tests have shown that the greater the car's state of disrepair, the greater the hydrocarbon emissions.

Benzene from gasoline is classified by the EPA as a Group A carcinogen for lung cancer. It has been shown to increase the risk for leukemia. Benzene levels are elevated in gasoline sold in cold climates, such as Alaska.

Volatile liquids, such as paints, charcoal starter fluid, cleaners, and kerosene for heaters and lamps, all add their fumes to the chemical soup that can be transferred to the house.

Transport Paths

Various studies have shown that the leakage through the HG interface can vary widely from 1% to 80% of total infiltration into the house. The Minnegasco study cited above found that the air leakage into the house through the garage averaged 24% of the total infiltration into the house from all sources. Transport paths via a furnace and air handler in the garage can be even more significant than the holes in the HG interface, since that air flow is driven by the fan in the air handler. If ducts in the garage are leaky, they will cause powered pressure imbalances, transporting garage air through leaky return ducts directly into the house or pressurizing the garage, increasing the pressure differential across the HG interface.

Yet even if the duct leaks are sealed,

“engineered” leaks are often designed into the air handling equipment, to enable the system to run correctly. The areas that allow combustion air to flow into the system, for example, cannot be sealed unless they are replaced by a dedicated source from outside the garage. If the access door is taped closed, this will make it more difficult to replace the filter and service the system. Some cooler air, either from the return ductwork or from the outside, is required to keep the control electronics from overheating. Such leaks need to be addressed by a qualified furnace technician.

Some Solutions

The most obvious solution to house infiltration from vehicles is to detach the garage from the house. Both the American Lung Association and Canada Mortgage and Housing Corporation (CMHC) state that for optimum IAQ conditions, the garage should be built as a structure separate from the house.

If the garage is a part of the house, sealing the HG interface is critical. This can be a particularly difficult task with a tuck-under garage, due to the indirect connections through the ceiling and the wall structures above. (For examples of tuck-under garages, see photo on p. 20.) Particular attention must be paid to air sealing details of any wall or ceiling penetration. Doors between the house and the garage must be treated with at least the same detailed attention as any other outside door.

Taking the air handler out of the garage will make it impossible for the fan to circulate polluted air throughout the house. If the air handler must be in the garage, the furnace should be physically separated from the garage—perhaps located in a sealed utility closet accessible only from the outside, and only sealed-combustion appliances should be used.

However, the house must be considered as a system. If the naturally aspirated furnace is located in the house, it may be dependent on infiltration from the garage to provide combustion air. Since energy-efficient gas furnaces allow very little waste heat to rise up the flue (relying on a pressure differential of as little as



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Can you spot all the potential hazards in this photograph of a typical North American garage?

4 Pa to draw the waste out), the furnace may be easily backdrafted by a bathroom fan, range hood, or clothes dryer—particularly if the HG interface has been carefully sealed. This is of greatest concern for retrofit improvements of the HG interface, particularly if the furnace has been operating properly.

An exhaust-only ventilation system lowers the pressure in the house and can actually increase the transfer of polluted air. However, any dedicated ventilation system in the house will dilute the pollutants drawn in from the garage, and will bring pollution levels down more quickly than would be the case if there were no ventilation system in the house.

One solution that is worth considering is to install an exhaust fan in the garage, which can create a negative pressure in the garage relative to the house, restricting or reversing the flow through the HG interface. The exhaust fan will also expel the polluted air to the outside. Both of these effects will be overcome when the much more powerful blower in an air handler located in the garage is running. CMHC has shown that if the garage is poorly sealed and drafty relative to the outside, an exhaust fan will have little effect.

The volume of flow for an optimum exhaust fan varies. If the garage and house are very tightly constructed, very little air flow is needed. A fan with a

10–20 CFM capacity may be adequate to render the pressure in the garage negative relative to the house. In most studies, fans of 150–250 CFM have been required. Care must be taken not to backdraft a furnace or water heater located in the garage as the garage exhaust fan competes with it for air. Since CO is very slightly lighter than air, the fan should be installed high on the wall or in the ceiling.

The exhaust fan must be allowed to run in the garage either continuously or on a control that will automatically start the fan when anyone enters the garage and will run for at least 20 minutes after a car has been brought in or taken out.

Homeowners must be told that a car should run in the garage for the shortest possible time. They must be told never to warm the car up in the garage, even with the garage door open. The garage should be ventilated after the car has been removed. Homeowners should never leave the door between the house and the garage open when the car is running. Remote car starters should never be left within the reach of children. With the new entertainment systems in cars, “just watching the end of the movie” can mean leaving the car running in the garage for half an hour. Turn it off and bring the movie into the house.

It would be safer to bring the cow back into the basement. At least the cow is a source of pollution that can be sensed

and would be dealt with. It is fortunate that most health problems caused by attached garages are limited to headaches or flulike symptoms. Builders have made the garage door a style statement on the front of many homes. Now they need to understand what living that close to the car really means. With the growing body of evidence supporting the conclusion that pollutants are drawn into the house from the garage, and the increasing volume of legal struggles, garages and CO are likely to be the next battleground for builders, homeowners, and renters, following in the footsteps of mold.



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